3.4 NEWTON-RAPHSON METHOD

The Newton-Raphson method is the best-known method of finding roots of a function $f(x)$. The method is simple and fast. One drawback of this method is that it uses the derivative $f'(x)$ of the function as well as the function $f(x)$ itself. Hence, the Newton-Raphson method is usable only in problems where $f'(x)$ can be readily computed. Newton-Raphson method is also called *Newton's method*. Here, again we assume that $f(x)$ is continuous and differentiable and the equation is known to have a solution near a given point. Figure 3.4 illustrates the procedure used in Newton-Raphson method. The solution process starts by selecting point x_1 as the first estimate of the solution. The second estimate x_2 is found by drawing the tangent line to $f(x)$ at the point $(x_1, f(x_1))$ and determining the intersection point of the tangent line with the x-axis. The next estimate x_3 is the intersection of the tangent line to $f(x)$ at the point $(x_2, f(x_2))$ with the x-axis, and so on. The slope, $f'(x_1)$, of the tangent at point $(x_1, f(x_1))$ is written as

$$f'(x_1) = \frac{f(x_1) - 0}{x_1 - x_2} \tag{3.9}$$

Rewriting Eq. (3.9) for x_2 gives

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \tag{3.10}$$

Equation (3.10) can be generalised for determining the next solution x_{i+1} from the current solution x_i as

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \tag{3.11}$$

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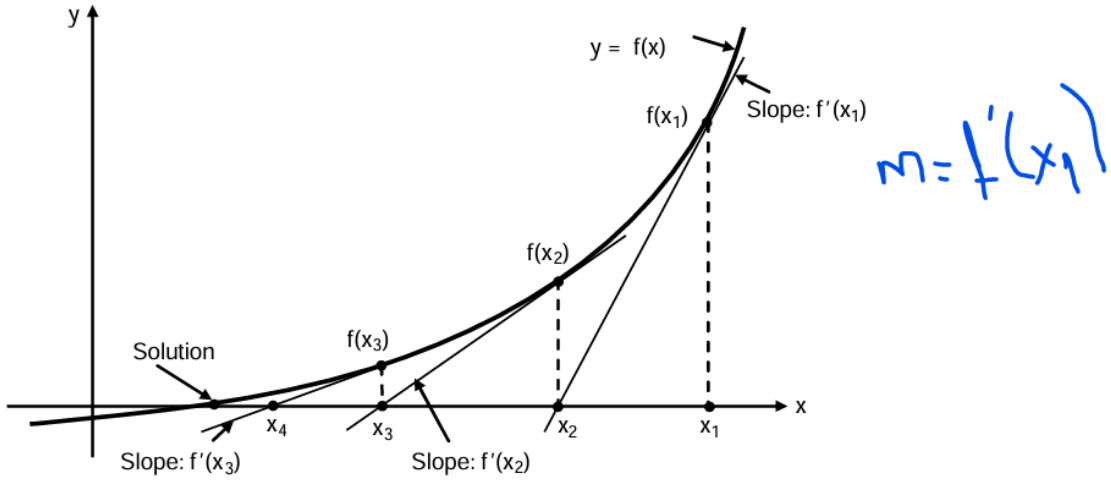


Fig. 3.4: Newton-Raphson method

So the formula fort he Newton-raphson method is this:

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ n=0,1,2, … (provided that $f'(x_n) \neq 0$)

Example: Use Newton-Raphson method to find the real root near 2 of the equation $x^4 - 11x + 8 = 0$ accurate to five decimal places.

Solution: First let us check that $x_0 = 2$ is a real root of $f(x) = x^4 - 11x + 8$.

f(2)=2^4-11(2)+8=16-22+8=2$\neq 0$, so that $x_0 = 2$ is not a root of f(x), now we will find approximation of the root using N-R method.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

For n=0

$$x_1 = x_0 - \frac{f(x_0)}{f\prime(x_0)}$$

Then $x_1 = 2 - \frac{f(2)}{f\prime(2)}$

f'(x)=$4x^3$-11 then f'(2)=$4(2)^3$-11=21

Then $x_1 = 2 - \frac{2}{21} = 1.90476$ so the first approximation for the root is $x_1 = 1.90476$.

Now let us find the second approximation,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f(x_1) = f(1.90476) = (1.90476)^4 - 11(1.90476) + 8$$

$f'(x_1) = f'(1.90476)$=4$(1.90476)^3$-11

If you substitute these values to

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

We find that $x_2 = 1.89209$

Then continue in this way

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.89209 - \frac{(1.89209)^4 - 11(1.89209) + 8}{4(1.89209)^3 - 11} = 1.89188$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 1.89188 - \frac{(1.89188)^4 - 11(1.89188) + 8}{4(1.89188)^3 - 11} = 1.89188$$

Why we stop here: We wanted to accurate to 5 decimal places. After x_3 the approximate root is same, repeat itself, this means that until this 5 decimal places, we found the root correctly, accurately.

SUCCESSİVE APPROXIMATION METHOD (FİXED POINT ITERATION)

Definition: The point x=a in the domain of a function g(x) is called a fixed point of g(x) is g(a)=a.

Example: Find all fixed points of $g(x)=x^2+x-9$ if it exists.

Solution: g(x)=x

$x^2+x-9=x$

$x^2-9=0$ then x=3, -3 are all fixed points of $g(x)=x^2+x-9$.

But not all function's fixed points can be found easily like this example:

Theorem: Assume that g(x) and g'(x) continous and differentible on a interval closed interval I=[a,b], $|g'(x)|<1$ on I and $g([a,b])\subseteq[a,b]$. Then g(x) has a unique fixed point on [a,b] and the iteration

$$x_{n+1} = g(x_n) \quad n=0,1,23....$$

Converges to this fixed point for any initial point $x_0 \in [a,b]$.

Example: Using the successive approximation method, find the fixed point of g(x)=1+0.5sin(x) to four decimal places.

Solution: $-1 \leq \sin(x) \leq 1$

Then $0.5 \leq g(x) \leq 1.5$

Hence $g(x): [0.5, 1.5] \rightarrow [0.5, 1.5]$

And g'(x)=0.5cos(x) then $|g'(x)| \leq \frac{1}{2} < 1.$

Therefore the iteration

$$x_{n+1} = g(x_n) \quad \text{n=0,1,23,..}$$

$$x_{n+1} = 1 + 0.5\sin(x_n) \quad \text{n=0, 1,2,3,...}$$

Converges to the fixed point of g(x) for any inital point $x_0 \in [0.5,1.5]$.

Let us choose the midpoint of [0.5, 1.5] for a initial point. So

Take $x_0 = 1$

$$x_1 = 1 + 0.5\sin(x_0) = 1 + 0.5\sin(1) = 1.4207$$

$$x_2 = 1 + 0.5\sin(x_1) = 1 + 0.5\sin(1.4207) = 1.4943$$

$$x_3 = 1 + 0.5\sin(x_2) = 1 + 0.5\sin(1.4943) = 1.4985$$

$$x_4 = 1 + 0.5\sin(x_3) = 1 + 0.5\sin(1.4985) = 1.4986$$

$$x_5 = 1 + 0.5\sin(x_4) = 1 + 0.5\sin(1.4986) = 1.4987$$

$$x_6 = 1 + 0.5\sin(x_5) = 1 + 0.5\sin(1.4987) = 1.49870$$

$$x_7 = 1 + 0.5\sin(x_7) = 1 + 0.5\sin(1.49870) = 1.49870$$

So the fixed point of g(x)=1+0.5sin(x) until four decimal places is x=1.4987.

Example: Find a real root of : $f(x)=x^3-2x-3$., correct to three decimal places using the successive approximation method (fixed point iteration).

Solution: First let us check that in which interval the root of $f(x)=x^3-2x-3$ exists using the intermediate value theorem.

$f(1)=1^3-2(1)-3=-4<0$

$f(2)=2^3-2(2)-3=1>0$

then f(x) has a real root in the interval between 1 and 2.

To apply fixed point iteration, we should find a function g(x) such that the fixed point of g(x) should be equal to the root of f(x).

We can choose the g(x) in many ways, but we should choose the one $|g'(x)|<1$.

$f(x)=x^3-2x-3=0$ then $x=\frac{x^3-3}{2}$ then take, $g(x)=\frac{x^3-3}{2}$. Then $g'(x)=\frac{3x^2}{2}$, $|g'(2)|=6>1$, on [1,2]. Then $g(x)=\frac{x^3-3}{2}$ is not a good choose.

$f(x)=x^3-2x-3=0$ then $x=\sqrt[3]{2x+3}$ then take $g(x)=\sqrt[3]{2x+3}$. Then $g'(x)=\frac{2}{\sqrt[3]{(2x+3)^2}}$, then $|g'(x)|<1$ for any $x \in [1,2]$.

Then we can use $g(x)=\sqrt[3]{2x+3}$. See that $g(x)\colon [1,2] \to [1,2]$

$$x_{n+1} = g(x_n) \quad \text{n=0,1,23,..}$$

$$x_{n+1} = \sqrt[3]{2x_n + 3} \quad \text{n=0, 1,2,3,…}$$

Choose the inital point $x_0 = 1 \in [1,2]$.

$$x_1 = \sqrt[3]{2x_0 + 3} = \sqrt[3]{2(1) + 3} = \sqrt[3]{5} = 1.25992$$

$$x_2 = \sqrt[3]{2x_1 + 3} = \sqrt[3]{2(1.25992) + 3} = 1.31229$$

$$x_3 = \sqrt[3]{2x_2 + 3} = \sqrt[3]{2(1.31229) + 3} = 1.32235$$

$$x_4 = \sqrt[3]{2x_3 + 3} = \sqrt[3]{2(1.32235) + 3} = 1.32427$$

$$x_5 = \sqrt[3]{2x_4 + 3} = \sqrt[3]{2(1.32427) + 3} = 1.32463$$

Hence the fixed point of g(x) is 1.324, therefore the real root of f(x) is 1.324 in the interval [1,2] correct to three decimal places. 5

secant method

The secant method is very similar to the Newton-Raphson method. The main disadvantage of the Newton-Raphson method is that the method requires the determination of the derivatives of the function at several points. Often, the calculation of these derivatives takes too much time. In some cases, a closed-form expression for $f'(x)$ may difficult to obtain or may not be available.

To remove this drawback of the Newton-Raphson method, the derivatives of the function being approximated by finite differences instead of being calculated analytically. In particular, the derivative $f'(x)$ is approximated by the backward difference

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \tag{3.17}$$

where x_i and x_{i-1} are two approximations to the root but does not require the condition $f(x_i) \cdot f(x_{i-1}) < 0$.

Formula fort he secant method:

$$x_{n+1} = x_n - f(x_n) \times \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}.$$

Notice that: in the Newton raphson method you start with one initial value, but in the secant method we need 2 inital points.

$$x_{n+1} = \frac{x_n f(x_n) - x_n f(x_{n-1}) - x_n f(x_n) + f(x_n) x_{n-1}}{f(x_n) - f(x_{n-1})} = \frac{-x_n f(x_{n-1}) + f(x_n) x_{n-1}}{f(x_n) - f(x_{n-1})}$$

Then $x_{n+1} = \frac{f(x_n) x_{n-1} - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$

Example E3.15

Determine a root of the equation $\sin x + 3 \cos x - 2 = 0$ using the secant method. The initial approximations x_0 and x_1 are 0 and 1.5.

Solution: $f(x)=\sin(x)+3\cos(x)-2$, $x_0=0$, $x_1=1.5$

Then

$$x_2 = \frac{f(x_1) x_0 - x_1 f(x_0)}{f(x_1) - f(x_0)} = \frac{f(1.5)0 - 1.5 f(0)}{f(1.5) - f(0)} = \frac{0 - 1.5(\sin(0) + 3\cos(0) - 2)}{(\sin(1.5) + 3\cos(1.5) - 2) - (\sin(0) + 3\cos(0) - 2)}$$

$$x_2 = 1.24488$$

Then next approximation, $x_1=1.5$, $x_2 = 1.24488$

$$x_3 = \frac{f(x_2)x_1 - x_2 f(x_1)}{f(x_2) - f(x_1)} = 1.21122$$

$$x_4 = \frac{f(x_3)x_2 - x_3 f(x_2)}{f(x_3) - f(x_2)} = 1.20788$$

Continue in this way

$x_5 = 1.20783 \quad x_6 = 1.20783$

İf the approximation repeat itself, you stop there. Here you are certain that until 5 decimal places your approximation fort he root is 1.20783.

1.2 Exercises

1. Find all fixed points of the following $g(x)$.
   (a) $\frac{3}{x}$ (b) $x^2 - 2x + 2$ (c) $x^2 - 4x + 2$
2. Find all fixed points of the following $g(x)$.
   (a) $\frac{x+6}{3x-2}$ (b) $\frac{8+2x}{2+x^2}$ (c) x^5
3. Show that 1, 2, and 3 are fixed points of the following $g(x)$.
   (a) $\frac{x^3 + x - 6}{6x - 10}$ (b) $\frac{6 + 6x^2 - x^3}{11}$

4. Show that $-1, 0$, and 1 are fixed points of the following $g(x)$.
   (a) $\frac{4x}{x^2+3}$ (b) $\frac{x^2 - 5x}{x^2 + x - 6}$
5. For which of the following $g(x)$ is $r = \sqrt{3}$ a fixed point?
   (a) $g(x) = \frac{x}{\sqrt{3}}$ (b) $g(x) = \frac{2x}{3} + \frac{1}{x}$ (c) $g(x) = x^2 - x$ (d) $g(x) = 1 + \frac{2}{x+1}$
6. For which of the following $g(x)$ is $r = \sqrt{5}$ a fixed point?
   (a) $g(x) = \frac{5+7x}{x+7}$ (b) $g(x) = \frac{10}{3x} + \frac{x}{3}$ (c) $g(x) = x^2 - 5$ (d) $g(x) = 1 + \frac{4}{x+1}$

1. Apply Fixed-Point Iteration to find the solution of each equation to eight correct decimal places. (a) $x^3 = 2x + 2$ (b) $e^x + x = 7$ (c) $e^x + \sin x = 4$.
2. Apply Fixed-Point Iteration to find the solution of each equation to eight correct decimal places. (a) $x^5 + x = 1$ (b) $\sin x = 6x + 5$ (c) $\ln x + x^2 = 3$

1. Apply two steps of Newton's Method with initial guess $x_0 = 0$. (a) $x^3 + x - 2 = 0$ (b) $x^4 - x^2 + x - 1 = 0$ (c) $x^2 - x - 1 = 0$

1. Apply two steps of the Secant Method to the following equations with initial guesses $x_0 = 1$ and $x_1 = 2$. (a) $x^3 = 2x + 2$ (b) $e^x + x = 7$ (c) $e^x + \sin x = 4$

6.1 Use simple fixed-point iteration to locate the root of

$$f(x) = 2\sin(\sqrt{x}) - x$$

Use an initial guess of $x_0 = 0.5$ and iterate until $\varepsilon_a \leq 0.01\%$.

6.2 Determine the highest real root of

$$f(x) = 2.1x^3 - 11.6x^2 + 17.5x - 6$$

(a) Graphically.
(b) Fixed-point iteration method (three iterations, $x_0 = 3$). Note: Make certain that you develop a solution that converges on the root.
(c) Newton-Raphson method (three iterations, $x_0 = 3$).

(d) Secant method (three iterations, $x_{-1} = 3$, $x_0 = 4$).